

DevOps Lab - Project 6

Name: Aryan Srivastava

PRN: 23070122055

AIM

Using Kubernetes, complete the features of application scalability by demonstrating how to create a cluster with the autoscale feature.

1. Code Implementation

deployment.yaml:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: social-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: social-app
  template:
    metadata:
      labels:
        app: social-app
    spec:
      containers:
        - name: social-app
          image: nginx
          resources:
            requests:
              cpu: 100m
```

2. Command Execution

Execution of: `kubectl apply -f deployment.yaml`

```
user@ubuntu:~/project$ kubectl apply -f deployment.yaml
deployment.apps/social-app created
```

Execution of: `kubectl autoscale deployment social-app --cpu-percent=50 --min=1 --max=10`

```
user@ubuntu:~/project$ kubectl autoscale deployment social-app --cpu-percent=50 --min=1
--max=10
horizontalpodautoscaler.autoscaling/social-app autoscaled
```

Execution of: `kubectl get hpa`

```
user@ubuntu:~/project$ kubectl get hpa
NAME          REFERENCE          TARGETS          MINPODS  MAXPODS  REPLICAS  AGE
social-app    Deployment/social-app  /50% 1          10       1        15s
```

Conclusion

Successfully implemented Social Media underlying infra challenges as required by the project specifications.